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University Social Responsibility of Higher Educational Institutions in the Global South: A Case of Indian Academia

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ABSTRACT

This study investigates higher education institutions' diverse social responsibility initiatives and assesses their impact on stakeholders (present students, future students, supporters, etc). Social responsibility emphasizes contributions to sustainable development and proactive solutions to societal and environmental challenges. Recognizing universities as crucial pillars of society, this research explores how they foster socially responsible values in students to cultivate responsible citizenship, focusing on the next generation of leaders. Data was collected using semi-structured interviews from 103 heads of STEM (Science, Technology, Engineering, Mathematics) departments in twenty engineering colleges in Delhi, India. A thematic content analysis using QDA Miner Lite software (ProvalisResearch) had been used to analyze the data. The results were organised into three categories. The study explores students' involvement in USR activities, implementation challenges, collaborations with corporations, NGOs, and universities, and energy-saving initiatives for sustainable campuses. It emphasizes promoting diverse student experiences, community engagement, and fostering professional and socially responsible development and provides recommendations for stakeholders to achieve environmental sustainability. Top of Form

KEYWORDS

USR (University Social Responsibility), HEI (Higher Education Institutions), CE (Community Engagement), Sustainability

Introduction

University Social Responsibility (USR) is the ability of higher education institutions (HEIs) to promote and apply a set of principles and values in management, teaching, research, and extension to address educational and social challenges in society. In the current globalized and competitive environment marked by economic crises, there is a critical need to explore new sources for economic and social growth and development. Social responsibility is an ethical framework signifying an entity's obligation, whether an organization or individual, to act for the overall benefit of society. It involves maintaining a balance between the economy and ecosystems. In this context, both individuals and organizations need to contribute to society and the environment. Recognizing that revenue is derived from these sources, it becomes a fundamental duty to give back. The United Nations established 17 Sustainable Development Goals (SDGs) as a global initiative for all countries to address climate change, eradicate poverty, and foster sustainability by 2030 [1]. Higher education is essential in achieving these goals, as universities play a pivotal role in generating knowledge and nurturing talent. The success of the SDGs relies heavily on significant contributions from university education and research. Higher education institutions, particularly universities, increasingly view social responsibility as an integral aspect of their mission [2]. They bear the responsibility of ensuring that academic knowledge directly contributes to improving human living

conditions and the development of civil society. So, they should focus on cultivating future leaders capable of initiating social change, transforming innovative ideas into practical steps for a more inclusive and sustainable future. [3]

Community Engagement (CE) is the higher education institution's counterpart to Corporate Social Responsibility (CSR). While teaching and research were traditionally considered core functions, CE is now widely accepted as the third core function. CE enables universities to leverage their expertise to meaningfully address the development challenges of their communities and country.

Community engagement is vital in higher education, equipping students for the workforce and instilling a sense of civic responsibility. It involves students engaging in conversations with communities and institutions to comprehend the lived experiences of their partners. This involvement enables Higher Education Institutions (HEIs) to have a tangible and positive impact on society at large (Driscoll, 2009).

Rapid industrialization, modernization, urban migration and population growth have incessantly depleted the non-renewable resources, resulting in scarcity. The exponential demand for limited resources has increased costs, exacerbating their scarcity. In the current globalized and competitive environment marked by economic crisis, there is a critical need to explore new economic and social growth and development sources. In this context, individuals and organizations need to contribute to society and the environment. The United Nations established 17 Sustainable Development Goals (SDGs) as a global initiative for all countries to address climate change, eradicate poverty, and foster sustainability by 2030. Science, technology, and engineering are vital to the achievement of these SDG, and engineers have changed the world, starting from the development of tools from stones to artificial intelligence and big data; they are needed in every sphere of society. Higher education in technical disciplines is also essential in achieving these goals, and universities play a pivotal role in generating knowledge and nurturing talents. The success of the SDGs relies heavily on significant contributions from university education and research. Top of Form Higher education institutions, particularly universities, consider social responsibility an integral part of their mission . They are responsible for ensuring that academic knowledge directly contributes to improving human living conditions and the development of civil society. So, they should focus on cultivating future leaders capable of initiating social change and transforming innovative ideas into practical steps for a more inclusive and sustainable future. Top of Form.

Currently, research on different types of CSR initiatives and the challenges faced by universities in developing nations is limited,. This study tries to fill this gap by shedding light on the experiences of the heads of departments of various engineering colleges in Delhi, India. Further, it also gives recommendations for the universities. The study includes the STEM leaders of only engineering colleges because technical disciplines are high in demand for innovation, and global warming is a cause of concern for the world. In this regard, green innovation management will help the university students (future entrepreneurs) adopt sustainable practices while innovating, which can reduce the negative environmental and economic impact. It now serves as a vital catalyst for national and regional economic growth, playing a pivotal role in fostering sustainable development .

Theoretical Background

There needs to be more literature specifically addressing social responsibility in higher education from Africa, excluding South Africa, as well as in Asia and the Middle East. Generally, the social responsibility literature complements the extensive body of work on community engagement. It offers insights into how social responsibility is manifested within Higher Education Institutions (HEIs) and their collaborations with communities and society. In a study of engineers in an HEI, which tried to analyze the student's contribution to society, it was analyzed that the inclusion of a course on sustainable development in the engineering curricula could motivate the students to contribute to society.. Engineering students must know the interaction between society, technology, and the environment so that the benefits of technology are positive for all. . In a study by an Australian university, the results revealed that to achieve sustainability and have a good quality of life, engineers need to create or modify existing technologies to enhance efficiency. For, e.g., they can develop technologies for the poor and the desolate to upgrade them in society and help them come out from the shackles of poverty. According to the world data, 12% of the population did not have electricity in their homes in 2016; in 2015, 2.1 billion people did not have access to safe drinking water. To address these challenges, the economic, social, human, and environmental impacts of engineering need to be considered. Courses that help engineers incorporate such value-based perspectives need to be added to the engineering curricula of educational institutions. . To make engineers sensitive to the demands of the global world, they need to be educated in courses like ethics and environment . Courses like STS (Science, Technology, and Society) have been developed to educate engineers and make them understand how their contributions can benefit society's welfare.

A study on Malaysian university students revealed that most were aware of the importance of environmental preservation but lacked exposure to actual social responsibility activities . In a study of Ethiopian HEIs, the USR activities were present, but the significant impediments were a lack of precise institutional arrangements, limited funding, inadequate collaboration/partnerships, lack of comprehensive data, and insufficient emphasis on morals, ethics, values, and principles . In another study carried out in the US, the results revealed that a collaboration between the government and the undergraduate students of a university on a sustainability project benefitted the government agency and the students. With collaborations, the students gain hands-on experience and also achieve recognition through the presentation of papers at conferences.. Recruiting students in firms as interns serves as a route for interaction between firms and universities.. Informal contacts, conferences and publications, consultancy, and recruitment from college were the factors that led to university-firm tech transfer. Education by a university on conducting sustainable research for its students is an important factor in determining the economic development of a country. .

Methodology

The aim of the study was to discuss the USR activities undertaken under the STEM leaders in different ways and how they can help to transform society in social and economic ways. The study utilized a qualitative explorative design rooted in hermeneutic phenomenology to gather descriptive data on the opinions and experiences of STEM leaders with USR activities in various departments of engineering colleges in Delhi, India. Interpretative phenomenological analysis emphasizes on providing detailed accounts of experiences in participants' own terms, avoiding predefined categories. IPA is considered phenomenological, focusing on interpretation, with a hermeneutic commitment to understanding . Top of FormData had been collected using semi-structured interviews with 103 Heads of departments of STEM (Science, technology, Engineering, Mathematics) disciplines, and thematic analysis has been conducted using QDA Miner Lite software (Provalis-Research). A census was done and emails were sent to all the

HoDs to give their nod to personal interview sessions where out of 127 STEM HoD's in a total of 20 engineering colleges of Delhi providing regular Btech degrees, a total of 47 agreed to give personal face-to-face interviews in the first round and 56 in the second round. So, a total of 103 leaders were involved, and their experiences with USR activities are the focus of this study. The interviews were conducted for a period of eight months in the respective departments of HEI's, starting from 01-10-2022 to 31-05-2023. An institutional approval letter was shown to the respondents to get their consent and they were informed that the results would be anonymised. The respondents included heads of the following ranks i.e, professor and associate professor. Open-ended questions were posed during interviews lasting an average of 20 to 30 minutes. While flexible, all respondents were asked a similar set of questions. The Consolidated Criteria for Reporting Qualitative Research, (COREQ) was completed (Appendix A). The first author conducted the interviews and no other individual was present in the interviews. Coding was done by both the authors. Both the authors had no prior relationships with the majority of research participants. However, the first author had met one participant at a conference. Research participants had received information sheets explaining the study's purpose without personal details about researchers. While participants knew the researchers' identities through email correspondence and could access their research interests online, this might have influenced their responses during interviews.

The thematic analysis approach was followed and it ensured that research findings were rooted in the understandings and experiences of respondents, reflecting their subjective meaning. Interviews were recorded, and data was coded and transcribed verbatim. The transcribed data were collected using The Framework Method and then analyzed thematically using coding process Data were initially organized into an Excel spreadsheet using a framework matrix. Each investigator coded the data. The thematic analysis involved constant data checking, and meetings to review and combine themes. The process focused on inductively identifying emerging themes rather than a hypothesis-deductive approach. After extracting the ideas, the coding had been carried out in three phases: (1)

Interview guide:

Q1 What are the types of USR activities that your department is involved in?

Q2 Do you face any challenges from the university about its operations?

Q3 Are there any linkages to develop the required skills among the students?

Q4 How do you inculcate the value of economic, social, or environmental responsibility among students, or can you provide any suggestions?

Note: Applied science, i.e., Physics, Chemistry, Mathematics, and Engineering branches: IT/Electronics/Mechanical/Biological Sciences/Civil/Design/Environmental/Architecture/Material Sciences/Energy Sciences/Textile and fiber.

Source: Prepared by the authors

Table 1: Information of the respondents:

Source: Made by the authors

Results

The qualitative data after coding has been categorized into the following themes:

Theme 1: Involvement in economic, social, and environmental activities

The qualitative data from respondents' interviews showed that in many departments of various universities, the USR activities were absent, or the Heads of the departments didn't take the initiative to conduct these activities, and only a few outreach programs were done to meet the institutional requirements. The average no. of projects ranged from 2-3 in a year. The universities that were engaged in USR activities had diverse perspectives. When asked about the type of USR activity their department is involved in, they were engaged in three types of activities: economic, social, and environmental. For e.g., Projects like Education for all, gender sensitization, mental health awareness, design for the marginalized, inclusive design, design for all, etc. Many were involved in social activities like woollen clothes distribution, blood donation, health camps, renewable energy, campaigns for saving water and electricity, plantation, anti-drug camps, cleanliness drive, voter's awareness, programs like adopt five villages nearby colleges, medicines to needy people, sleep awareness program for patients, education mentoring of govt. School girls in STEM, donation to Kerala relief fund, several projects done by the students targeting the benefit of the rural areas.

Some excerpts from the interviews are provided below.

P 8: "We address most of UN SDGs, particularly SDG 7, SDG 13, SDG 4, e.g.: nutrition application, etc."

P 69: " There are solar panels installed, and rainwater harvesting is also there in the university." Yes, we have 146 kW Solar energy rooftop in my college."

P 60: "Many of the faculty members in our department are working in the area of solar cell technology, green technology, sensors, cancer research, and targeted drug delivery."

P 83: "The faculty members are setting up the waste treatment plant."

P 72: "Some of my students work in corporate hospitals and provide free healthcare services to the poor every Friday afternoon, and they thank me for motivating them for this, and it is giving them more happiness than their salary."

P 2: "Our students participated in SAGY (Saansad Adarsh Gram Yojana) in a nearby village as a part of the NSS activity. They are also mentoring poor students and giving them free tuition".

P 81: "We are working on making the university zero waste. I have worked on composting kitchen waste to produce manure, refurbished furniture, and reduced condemnation. Reused and recycled civil and paper waste".

P 44: "Students have been involved in tree planting, park cleaning, waste recycling, training workshops, university fairs, etc.."

P 27: "Our students participate in many volunteering programs. They collaborate with NGOs for community outreach. I feel that such initiatives empower them with a sense of social responsibility and help foster an understanding of all the challenges of the society".Top of Form

P 91: "Under community services (CS), students actively engage in programs, volunteering in various homeless shelters, low-income schools, family support centers etc. The university also emphasizes providing students with different opportunities to bring about positive change in the local communities."

Sub Theme 1.1: Energy saving initiatives taken by universities

The participants had informed about the projects undertaken by their institutions and further information had been verified from the websites. DTU announced the upcoming installation of a 432-kW solar power plant, set to be installed on rooftops and covering an area of 7050 square meters . HMR Institute of

Technology & Management is a prominent private engineering college in New Delhi, India. It offers a range of courses in both engineering and management disciplines. In pursuit of a green vision, HMR Institute of Technology & Management partnered with Tata Power Solar to install a 146-kW rooftop system. The institute now operates entirely on green energy, meeting 100% of its energy needs through solar power. Additionally, students are actively involved in hands-on learning through access to live solar projects, fostering awareness and expertise in green technologies. The rooftop solar system at HMR Institute of Technology & Management generates 204,000 units of electricity annually. Over 25 years, it is expected to offset 3,679 tonnes of carbon emissions

Colleges under IP University have taken measures for energy conservation and usage of renewable energy to minimize the carbon footprint of the campus by leveraging solar power for electricity generation, implementing a biogas facility for renewable energy production, engaging in grid wheeling to optimize energy distribution, utilizing sensors for efficient energy conservation, incorporating LED bulbs and power-efficient equipment for reduced energy consumption, etc. Some more initiatives by the university in this regard include sustainable design with open courtyards, fly ash brick construction for energy efficiency, and energy-efficient screw chillers for air conditioning. LED luminaries, green reflect SOL glass windows, and clay-tiled surfaces enhance energy conservation. They also have provision for under-deck insulation, which reduces heating and air conditioning loads. A 140KW solar power plant generates 300,000 units annually, with surplus power fed into the grid. Solar water heaters are installed in four hostels, which provide 40,000 liters of warm water. Sensor-based electrical fittings are in key areas to minimize power consumption. Additionally, a biogas plant has been installed, which serves both lighting and educational purposes, emphasizing bio-resource recycling and societal needs Top of Form

The Green Policy at IIIT-Delhi Campus includes Yule e-bikes to reduce conventional fuel vehicular traffic, building orientation, and window design is there for maximum natural light utilization, steps to reduce air conditioning load are being taken, terrace and walls are equipped with insulation, dry cladding, and cavity walls are there to minimize the solar radiation, there is waste segregation into biodegradable, non-biodegradable, and e-waste, on-site composting and disposal through authorized agencies is available, biomedical waste from the health centre and labs are responsibly managed by the authorized agency (M/s Biotics) under DPCC, i.e., Delhi Pollution Control Committee

MAC INSERCH is a collective initiative at Maharaja Agrasen College, uniting students, teachers, and administrative staff in their shared commitment to environmental conservation, heritage preservation, and overall sustainable development. The initiatives include Top of Formfive functional rainwater harvesting pits, a 10,000-liter Effluent Treatment Plant that manages water discharge from washrooms, the canteen, and the girl's hostel. Energy-efficient measures include hermetically sealed windows and cavity-walled structures in air-conditioned rooms. In many areas, LED lights are used instead of CFL and mercury tube lights. The Cycling Club Cyclopaedia also encourages eco-friendly commuting with 20 practice cycles and over 12 high-end bicycles

MSIT has transitioned from incandescent bulbs to LED lights, replaced CRT monitors with energy-efficient TFT or LCD monitors, and advocates for efficient electrical systems. Regular maintenance of UPS batteries reduces charging current. There are frequent equipment checks which prevent excess electricity usage. The institute promotes energy conservation by urging staff and students to turn off fans and lights when they are not in use. Solar heaters in hostels reduce electrical power usage. There are transparent glass windows that bring natural light to classrooms. There will be a provision for solar-based LEDs on campus. The institution also adopts rainwater harvesting, where water is collected from an open terrace

and then directed to the ground through drains to sustain groundwater availability. Efforts for Carbon Neutrality include No burning of leaves or waste papers, leaves buried in soil; there is waste disposal in Nagar Nigam dumping yards, minimal AC usage, routine inspections for lower emissions, a vast tree cover neutralizes the carbon in the campus. In NSUT's electrical engineering department, a dedicated renewable energy lab will cater to Simulation-in-the-loop setups, Real-time simulators, DC microgrids, Wind energy emulators, and Fuel cell setups.

IIT Delhi has a supercomputer, i.e., Padum, which was inaugurated in 2015 and employs chilled water-cooling technology and innovative energy-saving measures, earning it a "green" designation. It ranked 28th on the Green 500 list of the world's most energy-efficient supercomputers. The ESPOB (Energy Storage Platform on Batteries Centre) is researching various energy storage technologies, including flow batteries, supercapacitors, fuel cells, electrolyzers, Li-ion, Na-ion, Mg-ion, Al-ion batteries, and Li Recovery. A 50 kg/day capacity pilot-scale reactor has been designed and is installed at the IIT Delhi campus to address the electronic and plastic waste streams Top of Form

IGDTU has installed solar water heaters in hostels and guest houses so that students and residents can use green and renewable energy. There are energy-efficient lights campus-wide, and five percent of the energy demand is being met by solar panels via Hero Future Energies Pt. Ltd. The campus also has carton boxes for segregating paper and plastic collection and also promotes the recycling of litter

GB Pant Institute of Technology had installed rooftop solar panels to harness solar energy, generate electricity, and reduce operating costs (G.B Pant Institute of Technology, n.d.).

The new building of Vivekananda College features a rainwater harvesting system that stores water for gardening and running tracks. PM2.5 aerosol concentrations are monitored using standardized equipment from the CSIR-National Physical Laboratory. The college promotes cleanliness through waste separation campaigns and provides green and blue dustbins throughout the campus. Organic waste is composted using compost pits and a compost machine. Medical and electronic waste are managed separately, with designated bins and storage areas. Laboratories handle non-hazardous materials, reducing the need for exceptional waste management. The well-ventilated classrooms minimize energy use and are aided by natural lighting and cross-ventilation. Solar panels and LED lights further reduce energy consumption. Garden waste is composted, and burning is prohibited. RO plants ensure safe drinking water, while a greenhouse supports plant replication and maintains greenery. Soundproof generators provide efficient backup power near Gate no. 2 (Vivekananda College, n.d.). HMR Institute of Technology and Management is now the first green and energy-efficient institute in GGSIPU after installing a 146 KW grid interactive solar power plant

Theme 2: Collaborations with the stakeholders

Most of the participants had a limited view of CSR, and they focussed only on the social dimension of it, whereas a few of the college heads had their students involved in the CSR activities through partnerships/collabs and had linkages with the industry (Microsoft, IBM, pharmaceutical companies, coding blocks) via MoU's and NGOs to conduct joint research projects. The respondents mentioned how training and placement cell helped sign MOUs to develop linkages with industries and how working with industry gave their students unique perspectives and an understanding of dynamic sustainability challenges and opportunities. Some excerpts from the interview are as follows:

P 22: "LinkedIn is a valuable platform for connecting professionals from various fields and bridging the gap between Industry and academia. Our students exchange ideas, insights, and experiences related to

sustainable practices in the supply chain. I think it is important for students to learn socially responsible entrepreneurship”.

P46: “The project involved collaboration with a technology manufacturer company, and the students gave the company a sustainable solution. They have taken this collaboration positively and seem to have a growing interest in partnering through the project course.”

P 91: “The college conducted a workshop, which around 150 entrepreneurs and business leaders participated in. It was focused on developing resilient business practices with a sustainability strategy.”

P 76: “A few of my students are engaged with the Tata Group on a project focusing on developing renewable energy for rural schools. The collaboration has led to a course redesign and the establishment of a co-curricular practicum”.

P11: “We have programs to train our students for collaborations with local industries. However, I feel that with the integration into the global knowledge economy, the students have a shift in their sense of purpose. I feel that the concept of public good is taking a backseat, and revenue generation and profit making are given more importance”.

P20: “The Placement Cell also helps us get many MOUs with the industry and conducts mandatory programs for students. Also, many of the students are hired as interns in companies to fulfil certain projects.”

Theme 3: Lack of resources and delays in conducting the CSR activities.

The challenges faced by the departments in conducting these activities were a shortage of funds and human resources, lack of support from administration (delays in approvals), poor communication, bureaucratic setup, and shortage of infrastructure. The participants also expressed concerns about low community engagement in social responsibility activities, the need for building local capacity, transparency issues, a narrower perception of social responsibility, a lack of consensus on implementing social responsibility initiatives, and cultural differences among students in a classroom also impacted productivity due to varying perspectives on fulfilling social responsibility. Apart from all these, many leaders raised concerns over environmental challenges like poverty and higher taxes as impediments to fulfilling social responsibility. A few excerpts of the interview are as follows:Top of Form

P 42: “Cultural diversity and indifference also pose a challenge to implementing social responsibility since we have a lot of foreign students here.”

P 73: “Lack of community participation hinders the proper implementation of social responsibility.”

P 13: “The universities prioritize cost concerns and competitive advantage over societal health, security, and ecological development. I don’t think they are willing to spend much on such activities”.Top of Form

P 98: “ Most of the students are doing projects with government funding that undergoes annual audits, and this leads to potential interruptions in project continuity. Since universities are constrained by stringent public sector accounting regulations that are basically aimed at preventing fraud and protecting stakeholders, we face challenges in project implementation, mainly in procurement and resource utilization.”

P 57: “Many times, I have helped my students get funding from my own funds, which had been allocated to me as an HOD for external uses, so that my students could go ahead with the project.” P 38: “Sometimes, the students need technical help for the project, and hiring professional or technical personnel is again a financial problem.”

P 89: "Diverse contexts and students from various backgrounds in a classroom pose a challenge to carry out a project as the foreign students might not feel a sense of belongingness to the people.

Theme 4: Different ways to inculcate social responsibility amongst students

The responsibility towards society was inculcated by the faculty in students through numerous ways like classroom learning, social practices (assigning a social project in co-project mentoring, expert lectures, conferences and workshops in operation with NGO's or mentoring by experts. Involvement of students in business clubs, eco clubs , rotary club, seminars, workshops, discussing real case studies in the classroom, involvement as drishti and NSS mentors and motivating them by becoming their models. Students were involved into various societies like ENACTUS, discussion of SDG's in the classroom, conducting awareness through workshops etc. Students also engaged in practical learning like arranging a visit to the slums to know the problems of the desolate and come up with low cost effective, innovative solutions for the same and in this way inequalities in the society could be reduced by uprooting the poor and letting them enjoy equal access to technologies. Community engagement had been emphasized by a lot of leaders. Many faculty said that they had certain courses designed to inculcate the values of social responsibility and some colleges had an additional paper of Human Values and Ethics.

Some excerpts from the interviews are as follows:

P 35: " Practical learning is needed. My students have installed a low cost water pump in the nearby slum to solve the problem of shortage of water. The students are also associated with many social organizations".We have rain water harvesting in our college. Apart from this,there are posters in toilets for saving water and in the classrooms for switching off lights when not in use".

P 47: "Activities outside the college are conducted like the sikh community in our college does donations, langar sewa especially in times of covid like situations or any other natural calamities".

P92: "We do design projects on socially relevant issues (maybe from education, healthcare, water and sanitation, sustainability, smart cities domains and so on). Hence, our students are trained to not just value but actually empathise and resolve the problems of stakeholders in the social environment".

P 8: "Biological Sciences students generated diverse health campaign materials for partner communities, addressing shared health concerns. These materials had been distributed as digital files and the partners got them printed and disseminated them within their respective communities.

The faculty also pointed out how they would know whether a student actually undergoes a change after engaging in a USR activity.

P 66: "I don't evaluate them but one can feel the change through discussions".

P 28: "When they became very knowledgeable and through expertise, I can feel that they have it in them now".

The faculty also made sure that the students had inherited the USR values in them by discussions, conducting counselling, assigning them marks in their projects for USR activities conducted by the students, classroom interaction, All round assessment test, viva, portfolio management, feedback forms, All round assesment test, viva, portfolio management, feedback from staff, quiz test, conducting presentations ,Pre and post test of how their business ideas have changed drastically after undergoing USR activities etc. Participants expressed how students are able to comprehend the urgency of climate change at the local and global level and via joint communications, they will seek collaborative solutions. The faculty also promoted social responsibility not only in education but in research as well. They inspired

the students to get real experiences and turn those community insights into research projects thereby returning the outcomes to benefit the community. This was done through projects, debates, and teamwork which sparked an interest in the real-world issues and cultivated the necessary skills in the students to address them. The faculty promoted a positive attitude toward volunteering and teamwork among students, familiarized them with diverse organizational environments through project participation and evaluated student's theoretical knowledge effectively.

P 09: "Partnering with day care centers, orphanages enables the students to understand team work".

Table: Approach of colleges towards fulfilment of SDG's

Source: Made by the authors

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lusion

Education must prioritize the integration of the 17 UN Sustainability Goals, with University Social Responsibility (USR) as a cornerstone reflecting ethical engagement for holistic development. Overcoming barriers such as resistance to change and financial constraints is essential for implementing sustainable practices. Despite challenges, many universities champion social responsibility in critical areas. Institutions should align with international benchmarks, forge strategic partnerships, and integrate USR into strategic planning to fortify their societal impact. Fostering student awareness and promoting University Social Responsibility (USR) is imperative for cultivating graduates with enhanced capabilities and a commitment to addressing societal challenges. There is a need for comprehensive approach to global sustainability. Universities need to enhance visibility of their actions and engage various stake

Limitation and Recommendation

The deliberate choice of qualitative study methods restricts the generalizability of the findings. Further, interviews included only the STEM leaders and not the rest of the faculty in a department. Our study suggests some specific recommendations as experienced by the STEM leaders.

Top of FormDiversified funding achieved through collaboration with local and global stakeholders is crucial for aligning educational efforts with industry and societal needs. Higher learning institutions should allocate budgetary provisions to prioritize implementing social responsibilities as a core activity in higher education. Partnerships/collaboration between universities and government/civil society/corporates can yield additional funding and research opportunities for USR activities. The government can mandate legislation and enforce strategies to ensure that HEIs prioritize and implement social responsibility to the public. University leaders can drive Social Responsibility (SR) by incorporating it into organizational strategy, creating dedicated units and roles, and aligning rewards (compensation, awards for impactful teaching and research) and performance evaluation measures (volunteering, advisory services). This establishes an essential infrastructure for collaboration with external stakeholders and internal legitimacy for social responsibility.

The universities should engage in Service-learning. It integrates learning objectives with community service to offer a practical and progressive educational experience that addresses societal needs. E.g., courses, mentoring programs, student initiatives, and internships. Top of Form Service-Learning and Civic Engagement aim for the common good, offering exceptional opportunities for civil society, students, and lecturers to apply academic knowledge to address social needs within a community context. Integrating

various Service-Learning formats into the curriculum ensures institutional sustainability and fosters the acknowledgment of social responsibility in higher education. Top of Form Top of Form Universities should incorporate the SDGs (Sustainable Development Goals) into their teaching and learning by adding undergraduate courses which address them and make sure that this learning makes a difference in the world. They shouldn't have a motto to adopt it to improve their university rankings but through these reforms instill a sense of social responsibility not only in academic work but also in other functions that link universities with real-world problems. Sustainability rankings, accreditation standards emphasizing social impact, and funding instruments for sustainable development significantly shape organizational practices for social responsibility and impact assessment in universities. These mechanisms create benchmarks, standards, and financial incentives that drive institutions to prioritize and enhance their commitment to sustainable and socially responsible practices.

References

Women in STEM workforce: A review from Southeast Asian Economies

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ABSTRACT

Science and technology has been considered a male-dominated profession with women holding fewer leadership positions. They are universally under-represented in STEM (Science, Technology, Engineering, Maths) fields. Retaining women in education and, hence, careers in science is essential for an economy's national development. They can bring a different outlook for every sphere by making it more diverse and leading to women's empowerment. A review of the literature has been done to determine the participation of women in the STEM workforce with a particular focus on Southeast Asian economies, which mainly comprise of the developing nations. The study identifies the reasons for women not being able to climb the hierarchical ladder in scientific careers. Furthermore, the initiatives taken by these nations to increase the participation of women in the STEM workforce have been discussed, followed by Interventions by the United Nations on this pertinent issue.

KEYWORDS

Gender Studies, Leadership in STEM, Southeast Asian studies

1. Introduction

For an economy to progress, Science, technology & innovation (STI), and human development are crucial, women represent almost half of the global population, and they can contribute a major strength to the socioeconomic development of a nation. Top of Form They are a vital human resource and pillars of the world economy. They have contributed in almost every sphere of life by being powerful agents of change. They are even leading the country (women president, speaker, etc), being a source of inspiration to many. In this context, the far-reaching benefits of diversity and gender parity in leadership and decision-making are increasingly being recognized in all spheres. Women as leaders and decision makers ultimately play a critical role in advancing gender equality and furthering the nation's economic, social and political progress. Therefore, their contribution should not be ignored, and they must be represented in decision-making in businesses, educational institutions, communities, and politics not because of gender parity but because of their merit. Sadly, the S&T; sector, which is considered a key element in the economic growth of a nation, is heavily male-skewed. Traditionally, it has remained a male-dominated area in many parts of the world where men still occupy the top positions. The presence of women in higher positions of power in this sector is still an exception. The growth in the fraction of women in S&T; has been slow for many decade We need to understand that building scientific capacity is a shared responsibility and, in this regard, a serious concern has been raised since the 1970's for the inclusion of women and gender parity in S&T; under the auspices of UN seminars, conferences being organised nationally and internationally. Delegates drafted the UNDP Beijing Express Declaration for advancing women's equality and opportunities Top of Format the Fourth World Conference on Women, Beijing, China, in 1995 where

women's empowerment was at the forefront. Since then, policymakers and planners have recognised the importance of women in leadership positions and efforts are being made. Developing nations that have made commendable progress have done it primarily through empowering women. So, the full involvement of women in scientific and technological endeavors is essential for rapid national development and the sustainable happiness of the people. Because of the persistent efforts by the governments of different nations, the number of women in leadership positions is on the rise from a local to a global level.

Despite some progress, women scientists are paid less, promoted less frequently, and more likely to quit research careers than men of similar qualifications (Inter-Academy Panel on 'Women in Science in India', 2016). Inclusivity and diversity in leadership positions are important as they bring a fresh perspective to decision-making. Empowering professional women and women at the rural and grassroots levels of developing countries is necessary. Women in rural areas need to be valued as a precious human resource and not a liability as they can change the social stigmas of the society by their upliftment through science and technology, thereby, questioning the societal norms of the patriarchal mindset where head of the family is generally perceived to be a male, relationship between work and family which act as hurdles for women in science. They even join the labour market, but their number dwindles as they move up at the organisational ladder. In the engineering sector in many nations, the situation is even worse, where the participation of women in education and employment is low.

2. Review of Literature

Articles were searched from the Google Scholar database with the following search terms used; 'Women in STEM workforce', 'Barriers to women in STEM', 'Women in STEM workforce in South East Asia', 'Gender inequality in STEM professions in South East Asian economies', etc. Separate searches for this particular topic have been done for the ten members of ASEAN, i.e., Brunei, Cambodia, Indonesia, Laos, Malaysia, Myanmar, the Philippines, Singapore, Thailand, and Vietnam.

2.1 Participation of Women in the STEM workforce

* Historically, women need to be more adequately represented in Science, Technology, Engineering, and Mathematics (STEM) education, leading to a corresponding underrepresentation in the STEM workforce. As per a 2017 report, women account for just 30% of STEM students in higher education worldwide. The absence of women in STEM organizations may lead to a lack of diverse perspectives, hindering the generation of innovative ideas. Homogeneous workforce in STEM are less likely to navigate effectively in dynamic and varied environments. Consequently, advocating for educational and career opportunities for women is viewed as a strategy to tap into the full potential of the entire pool of STEM innovators. While Western countries have progressed towards more egalitarian social structures, many Asian countries still largely adhere to patriarchal social value systems. Globally, efforts are underway to boost women's involvement in STEM education, civic engagement, and the STEM workforce. Some Asian countries have witnessed significant strides, with women notably increasing their presence in specific STEM fields, such as computer science and computer engineering. Historically, educational opportunities, especially in STEM fields, have favored males over females. While Western countries have shifted towards more egalitarian social structures, many Asian countries still maintain predominantly patriarchal social value systems. In the face of gendered challenges, numerous Asian women persist in pursuing STEM careers, navigating male-dominated work environments daily. Despite increased representation in specific STEM fields, women in many Asian countries often encounter limited access to leadership positions. Excluding women from STEM organizations risks overlooking valuable innovative ideas. Homogeneous workforces struggle in dynamic, diverse environments. Thus, promoting educational and

career opportunities for women is seen as essential to harness the potential of STEM innovators fully Top of Form

In India, participation of women up to Ph.D. is about 20-25%, but in the positions of Professors or Heads of departments/divisions, it falls to 10-12% (Inter-Academy Panel on 'Women in Science in India', 2016). Because of the trickle-down effect of few role models for women engineering students, there is a dearth of academic women engineers in visible leadership positions, which is responsible for their under-representation in academia, industry, and government. We need women in science at every level. According to a study on women in leadership, women spend more time contributing to diversity, equity, and inclusion efforts, 07% of men as compared with 11% of women (Forbes, 2021). Encouraging women faculty to consider formal leadership positions and providing structured opportunities to explore those options will likely result in those faculty members' participation as academic leaders). Working women engineers have found less stress in managing work-life balance, and the gender parity index for women in science is more than one. Very few numbers of women engineers are in the decision-making positions. The employment of women engineers has increased, but they are primarily concentrated in the lower levels because they feel that the workplace is very masculine and they do not feel very comfortable in an organisation.

* 2.2 Reasons for women remaining in the lower levels of hierarchy in scientific fields

Women scientists face innate prejudice and bias in the workplace. Patriarchal norms contribute to gender discrimination challenges. There is widespread gender insensitivity, which exacerbates workplace issues, and explicit sexual harassment remains a pervasive problem. Gender is embedded in the socio-cultural rules of a workplace, and certain practices are biased towards men, which hinders the chances of women to get to the top.. Women perceive themselves to be more fit in the managerial roles unlike men who consider themselves apt for technical roles because of gender stereotypes . In the case of CSIR laboratories, despite a significant contribution by women scientists in the output, they have not been able to reach the position of a head because of lack of visibility in the organization. They do not move up the ladder at the same place as men despite having the same experience as their male colleagues.. A possible reason for this is the Indian culture, which requires women to devote time to family, thereby making them less productive in the perception of the males working in research organizations. Also, men remain in the upper posts because sexism exists in the work environment and is ignored by both men and women. . It was observed that women were more coordinated and the ones who shared information with all. They were satiated with their current position with low expectations to move up the ladder, unlike optimistic men. Another reason why women scientists are at the lowest levels is the inability to protest against wrongdoing and the lack of motivation. . Unconscious gender bias existed in institutions when women who were well-accomplished asked for laboratory space, financial resources, and awards to be granted based on merit. . Women scientists are rarely appointed as the chairman of various committees in the research institutes and are very few at the top level of the hierarchy. . They are smaller because of the existence of patrifocal social norms.. Age could be a reason for women in the middle ranks being more productive in research than men because of less burden with grown-up children, and it tends to be even higher after they reach 50's as now they are free to devote time to research.. Because of the belief that women face a dual burden of work and homely responsibilities, they are considered less productive in the workplace. This could be because of the total time devoted, which could be reduced due to family responsibilities or staying back at odd hours, etc. The younger male scientists did support their wives, who were also scientists, in sharing the household responsibilities and taking care of the family, but they did accept that despite their changes in thinking, the stereotypes still existed at the workplace. Apart from this,

the social and cultural background of women determines their entry into science. The more educated parents with higher incomes encourage women's entry into male-dominated fields like science, but in the later stages of their careers, it does not affect them in any way.. Many women researchers in STEM distance themselves from other women in the workplace whom they feel are overreactive to gender bias issues or who counter them because these women feel that because of countering the conflicts, it rather leads to their exacerbation and not the resolution of those issues. In India, unlike other nations, there is a belief that women are equally capable of doing science. In fact, if we look at the no. of publications, women are not behind men, but in spite of this, the workplace is gendered.. In scientific research organizations, the socio-cultural context plays a crucial role in determining the level of gender bias. Because women are unable to socialize and form informal contacts with those in power, they are discriminated against when it comes to their promotions. This has been evident mainly in the case of junior women scientists. . Women in academic science are less than men in terms of faculty. There are more women than men in the position of assistant professor, but very few in the post of professors as compared to men as women are considered warmer and more social, whereas men are considered competent and innovative. Because of lower informal networking, women are less engaged in innovation activities . They are supposed to maintain their professional identity at the workplace and keep their one aside as it can hamper their professional image as one-woman engineers seem to represent all woman engineers. . Patronizing attitudes of men in the workplace act as a hindrance to the progression of women in terms of collaboration and networking.. It is mainly in academia that females get stuck under one specific male chauvinist head, and there is no option of switching. Interorganisational networks can help women scientists and support them to escape the hierarchical setup. . Casual odd-hour meetings, which are unproductive according to women scientists, provide an edge to men in having informal interaction with the ones in power. . It is hard for women in academia to contact industry scientists and file patents. . Informal communication is prevalent in the departments of S&T; universities. However, it is a matter of the only boy's club, which gives them an edge regarding networking inside and outside the committee. Socialization is a problem because of the culture of segregation of men and women in the workplace . Gender segregation in Turkey's R&D; sector is less than any other nation like the US because a boost for women's participation in natural sciences and engineering fields via public education and a gender-neutral university entrance exam has led to a surge in women innovators. . Apart from the formal and informal structural constraints, lower funding grants to women compared to men are another reason for their higher attrition rates. They have to depend mainly on personal sources for access to financial capital.. Grant funding is important for a scientist as it enhances productivity in the sense that a researcher can use it for training students or staff working under it on a project; it helps expand opportunities to collect data and also in forming more networks or collaboration. . Sadly, gender discrimination exists even in grant funding because of the stereotypes.

2.3 Status of women in STEM in Southeast Asian nations

ASEAN aims to foster economic and security cooperation among its ten members: Brunei, Cambodia, Indonesia, Laos, Malaysia, Myanmar, the Philippines, Singapore, Thailand, and Vietnam. Girls in primary and secondary education in numerous ASEAN Member States (AMS) outperform boys in mathematics and science, but this academic advantage does not always result in their pursuing further STEM education or careers. Although more women enroll in higher education in Southeast Asia than men, they are significantly underrepresented in STEM disciplines. Only one in six STEM students are women, highlighting a pronounced gender gap in tertiary education Top of Form. Although women constitute 39 percent of technology majors, a lower percentage compared to the 56 percent seen in other fields of study,

their representation in the region's technology sector is 32 percent. This contrasts their overall presence in the workforce, where women make up 38 percent

Table 1: Percentage of Female Tertiary Education Graduates by Field (WEF 2022)

Source: World Economic Forum 2022

In recent years, Singapore has witnessed heightened efforts from individuals, social organizations, institutions, and private companies advocating for increased representation of women in STEM (science, technology, engineering, and mathematics) fields. During 2014–2016, various local events aimed to promote women in STEM fields, including STEM Week by the National University of Singapore, STEM Workshop for Women by the Singapore University of Technology and Design, a UN Women's Committee panel discussion, and the Women in Engineering, Science, and Technology symposium by Nanyang Technological University. These initiatives respond to the need to expand the labor force, addressing challenges posed by a shrinking and aging population, as per the Ministry of Manpower (2016). Encouraging women's participation was a strategy to mitigate this issue. In Singapore, while the overall gender ratio is nearly equal, the representation of females in engineering programs is less than 20%. The University of Hong Kong (HKU) leads in gender representation with 28.9% of women leaders, predominantly from non-STEM backgrounds, primarily at the departmental level. HKU also excels with 14.3% female institutional leaders. National Taiwan University follows closely, boasting a notable 24.6% representation of women in leadership, showcasing significant progress in gender equality and university management among East Asian world-class universities. Private institutions dominate higher education in Asia (e.g., Japan, South Korea) with up to 80% student enrolment, whereas Latin America, which was historically tied to Catholic-founded private universities, now sees a shift towards for-profit institutions, indicating a growing preference for profit-driven education. Cambodia is amidst an economic transformation amidst the global Industrial Revolution 4.0, creating a heightened demand for STEM graduates. However, the nation encounters challenges in attracting students to STEM fields in higher education. Addressing this requires initiatives not only at the classroom level but also through extracurricular activities in upper secondary schools and broader societal efforts. Top of Form. Cambodian private university academics note the absence of research, citing no institutional requirements. Public universities show limited, locally published research, often diminishing post-masters thesis completion.

Globally, there is a significant gender disparity among researchers, with men outnumbering women by a ratio of 71:29%. Interestingly, countries with lower research and development expenditures, such as the Philippines and Thailand, have a higher proportion of female researchers, exceeding 45% (UNESCO, 2012). Both personal and environmental factors influence academic research engagement and productivity. Top of Form. Top of Form. A conducive environment grants access to essential tools and resources, enhancing productivity. Management support in Malaysian public universities correlates with increased research engagement.

The Malaysian higher education landscape has transformed since the 1990s, notably in the public and private sectors. However, persistent gender stereotypes contribute to a notable imbalance among senior staff in public institutions. The private sector has seen slow progress in addressing this gender disparity over the past two decades. A Malaysian private university ranked second nationally in the Times Higher Education Ranking 2021, which prompted an exploration of how academics in private institutions connect to Knowledge Sharing (KS) behavior and subsequent research engagement. In a Saudi Arabian public university, 39% of faculty acknowledged engagement in intra-institute research collaborations, but cultural identity emphasizing Islamic values hinders strong motivation for collaborative research with colleagues.

As a result, faculty predominantly pursue individual research endeavors. In Thailand, women are significantly represented in research across diverse sectors and fields. Within government, higher education institutes, public enterprises, and private nonprofit organizations, they constitute a majority, accounting for 54.2% of researchers. Meanwhile, in the industry sector, women still make up a substantial portion, comprising 47.5% of researchers. In Myanmar and the Philippines, women are the predominant proportion of science researchers compared to men, as shown in the figure below. Women hold 5.7 percent of the highest research positions (typically director of research or full professor) in Brunei Darussalam, where representation of women decreases with seniority, which is in line with the global trend. Myanmar is the exception, where women hold 71.6 percent of the highest research positions.

Figure 1: Gender Composition of Science Researchers

Source: UNESCO 2020

Brunei's government has prioritized science education, allocating sufficient resources across all schools. Despite well-equipped government schools, there is a concerning trend of declining enrolment in the science stream. Additionally, examination results suggest unsatisfactory outcomes, highlighting the need for further attention to enhance science education effectiveness. Gender inequality in science education is evident in Bruneian textbooks. An analysis of textbook content reveals disparities in the representation of males and females in illustrations and text. This examination paints a picture of how science is portrayed, indicating the extent to which it is presented as a pursuit equally suitable for both boys and girls.

In the Brunei civil service, women constitute about 50.4%, with a higher presence in Divisions II, III, and IV. Despite gender gaps in academic achievements favoring women, they remain underrepresented in senior-level civil service positions, where only one in five holds such a position. In STEM fields, female tertiary graduates in Brunei reached 49%, surpassing rates in Singapore and the United States between 2014-2016. However, historical trends show a preference for 'lower level, clerical jobs' among Bruneian women. In Myanmar, a notable gender disparity exists in leadership positions, predominantly filled by males. While there may be some women in senior roles, this presence does not equate to gender equality. Widespread inequalities persist, encompassing areas such as political representation, societal perceptions, gender stereotypes, and access to resources.

While career advancement in Indonesian academia is theoretically nationally regulated to ensure equal opportunities, women in Islamic HEI (Higher Education Institutions) face disparities. They exhibit lower productivity in publications, hold lower academic ranks and leadership positions, and earn significantly less than their male counterparts. The suggestion is for these institutions to specifically empower women by supporting their continuing education, professional development, and career opportunities. In the UNDP's 2010 Human Development Index, Indonesia ranks 124th out of 187 countries, lower than Brunei (33) and Malaysia (61) but higher than the Philippines (112). Notably, Indonesia outperforms Vietnam (128) and India (134). Despite its overall ranking, Indonesia excels in expected years of schooling (13.2), surpassing most of these countries, except Brunei (14.1), and outperforming Malaysia (12.6). In a study of higher education institutions in Catbalogan City, Philippines, 30 participants shared their experiences through a validated questionnaire. The findings revealed that women respondents encountered gender biases in the workplace, particularly within the institution's regulations and procedures, despite government initiatives promoting gender equality.

. In many higher education institutions in the Philippines, women mostly occupy middle and senior-level positions but are less represented in the highest-ranking positions . In traditional times in Thailand, the presence of a bilateral and matrilineal kinship system and the high work participation rates among women in the modern, rapidly globalizing economy were sometimes seen as indications of minimal gender discrimination. In the 1990s, the demand for women's studies programs emerged, with both successes and setbacks. Graduates from these programs became leaders and have since played crucial roles in defending women's rights. A notable divide exists in Thailand between the household, where women play pivotal roles in family sustenance, and the public political culture. In the latter, women's internationally renowned 'pleasing personality' is a crucial element in a complex structure that often reflects subservience . The 2007-2009 Thai Labor Force Surveys indicate that only one-third of employed STEM graduates work in STEM occupations. While STEM degree holders in STEM fields enjoy a 30% wage premium, those working in non-STEM fields experience only a 2% premium. This suggests that STEM education yields significant returns in STEM occupations but lower, or even negative, returns in non-STEM fields, particularly for women . The International Labour Organization (ILO) estimates that 44% of employment, equivalent to over 17 million jobs, is at risk of automation in Thailand. Women are disproportionately employed in jobs with low STEM skills, making them more susceptible to automation. They are 50% more likely than men to face job loss due to automation. In the industry, 85% of total female labor holds low-skilled occupations. The ILO program aims to enhance women's acquisition of critical soft and technical STEM-related skills, addressing skills mismatches to boost productivity and enhance competitiveness in the evolving landscape

Top of Form Table 2: Average Monthly Earnings of Employees in Professional, Scientific, and Technical Activities by gender (in US \$)

Source: International Labour Organisation

Across the professional, scientific, and technical sectors, women generally hold lower representation than men, except Myanmar. There persists a significant gender pay gap. On average, women earn as little as 68 percent of what men earn in these sectors, highlighting substantial disparities in earnings based on gender. Women in STEM leadership across ASEAN are underrepresented but generally outperform global averages. In STEM policy roles, women are still underrepresented across all government levels. For example 2017, women constituted 35% of permanent staff positions in Lao PDR's Ministry of Science and Technology . In Malaysia, women occupied 23.7% of board seats in life sciences and healthcare companies in 2021, surpassing the global average of 21.3%. In technology, media, and communications companies, women held 7.8% of board seats in Indonesia, 20% in Malaysia, and 22% in Singapore in the same year . Women are underrepresented in executive positions, with only 14% holding chief technology officer roles in Singapore and 12% in other ASEAN Member States (AMS). Additionally, companies are recruiting women from Asia-Pacific offices for global roles based elsewhere

2.4 Barriers to women's participation in STEM institutions in Southeast Asia

Women encounter various barriers in pursuing, staying in, and progressing within STEM professions, ranging from personal and cultural hurdles to structural and systemic obstacles. These challenges lead to low representation of women in STEM fields. Women's active involvement in STEM careers, whether as entrepreneurs, employees, researchers, or leaders, is vital for fostering innovation and driving sustainable economic growth across the ASEAN region.

Math achievement was connected with coping mechanisms, study strategies, learning styles, and self-efficacy in a study of Year 13 students in Brunei. Poor arithmetic performance prevents students of

both sexes from enrolling in science and technology programs, which are essential to Brunei's industrialization. Particularly, fewer men than women are admitted to Higher Education Institutions (HEIs), which may be related to lower male achievement rates in GCE "A-level" mathematics. There are apparent differences in the ideals of Asian and Western societies. Asian communities encourage collectivism, which includes asking for help from peers, whereas Western cultures emphasize independence. In addition to motivation, socioeconomic situation impacts academic achievement in mathematics and other courses. This element is commonly mentioned in talks on educational outcomes (Mundia & Metussin, 2019).

A self-rating questionnaire survey involving 1281 first-year students across eight higher education institutions in Cambodia explored factors influencing their initial choice of science and engineering major. Findings indicate that individual factors, encompassing personal characteristics and attitude, had the most significant impact compared to family and high school factors. Intrinsic motivation, notably, is a crucial factor empowering academics to excel in their research pursuits.

Women are more likely to major in STEM and computer technology outside the capital in Cambodia. Markers of urbanization and socioeconomic development largely predict women's participation in health and IT majors. It is interesting to note that there is a negative correlation in these industries between gender equality traits and women's representation.

Female students' enrollment in science classes fell precipitously from 97% to 42% in Cambodian upper secondary schools between 2014 and 2019, whereas their enrollment in social science programs rose from 4% to 58%. The easiness of national exams, high passing rates, pursuing grades, attitudes toward science, and the cost of private tuition were all significant influencing variables. Female students stated they had no intention of pursuing STEM degrees, frequently due to family disapproval of science-related fields. Top of Form. Social Cognition Career Theory (SCCT), which was developed from Albert Bandura's social cognitive theory, was deemed the most relevant framework for analyzing the situation of women in STEM in Cambodia. It looks at how societal influences, gender, cultural norms, genetic characteristics, and unforeseen life events can affect one's career choice. It also highlights how an individual's job decisions are shaped by the interaction of their self-efficacy, outcome expectations, and personal aspirations.

Deeply rooted social conventions, religion, and culture all play a part in Myanmar's widespread inequality. The heritage of patriarchy and authoritarianism sustains the acceptance of women as a secondary gender. Women's equality is limited in this country due to the macho state and male-dominated institutions; national participation in upper and lower houses is less than 14% (Khin Khin Mr, 2021). In Myanmar, women's and girls' job advancement is impeded by institutional restrictions. Elevated academic barriers, especially in fields like engineering and medicine, disproportionately benefit men, restricting girls' educational opportunities and violating their human rights. This framework prevents women from achieving leadership positions, perpetuating gender inequality (Ei Yin Yin Phyu, 2023).

While there was initial progress, Myanmar's existing political and socio-economic challenges demand extra research and dedicated efforts to promote quality education and establish gender justice. The term "culture" in Indonesia has to consider the complexity brought about by several religions, especially Islam, which is the predominant religion and cannot be limited to the varied local ethnic groups. In this environment, the concept of "local culture" becomes complex. In Indonesia, women who want to go to college must negotiate several public spaces in addition to intricate cultural and religious norms. It might be not easy to juggle responsibilities as workers, instructors, spouses, mothers, and students. These women must manage a complex network of social, cultural, and familial roles, reflected in their juggling

these varied responsibilities. Top of Form. Even though gender equality is becoming more widely acknowledged in Indonesia, women in postsecondary education face obstacles in pursuing senior leadership roles. The leading causes of obstacles are not institutional barriers but family obligations, patriarchal cultural norms, and inadequate assistance Top of Form

Female scientists from Western Europe and North America said Singapore had a less family-friendly work environment in qualitative research with STEM professionals. On the other hand, Asian participants mentioned that they could focus more effectively on their scientific careers if they had domestic servants. Male secondary students in Singapore showed a predisposition for engineering and mathematics programs, while female students tended to gravitate toward the arts, social sciences, and medicine. Students who identified as female were more inclined to think that their skills were necessary. Males preferred engineering in junior college, while females selected the arts and social sciences. The latter group's choice was influenced by professional goals and parental guidance (Teo & Yeo, 2017).

According to a case study on women's careers in agricultural science in the Philippines, women who have two jobs have more significant difficulties than males. Delays in training, moves brought about by husbands' jobs, and restricted mobility are some of the restrictions. Gender roles continue to be stereotyped in culture, which affects interactions at work and prevents women from rising to top positions. Cultural obstacles remain the principal obstacle preventing women from advancing in their careers, even in emerging countries where the number of women working in research institutions is rising . According to a study conducted on 4967 junior high school students in Vietnam, gender has no bearing on academic success in STEM fields. Still, female kids did better when their parents worked in non-manual jobs than male students. Improved study scores were connected with non-manual jobs, higher parental education, and better economic standing. On the other hand, poorer academic achievement was linked to having 0 or more siblings. These results highlight the significance of preserving family size and giving women access to education to guarantee fair resource allocation for children Top of Form. In a qualitative study that included 21 participants in a Vietnamese university, the women leaders were inspired to progress in their careers by ongoing support from family and mentors, shifting employer perceptions, and their proven leadership abilities.

In the Lao PDR, economic priorities and educational goals sometimes collide, especially in underprivileged areas where kids must help support the family. Primary education quality, the competency of teachers and principals, the absence of learning resources, multi-grade teaching, and a poor pedagogical support system are among the issues that need to be addressed. These limitations greatly influence how girls and women will participate in STEM education in the future, both in their career trajectories and at higher levels Improving the quality of life for women in Timor-Leste requires meeting their fundamental requirements, which include access to primary education, health care, food, clothes, and shelter. These needs are frequently the focus of traditional aid. To overcome women's adverse conditions relative to males, it is equally vital to fulfill their strategic demands. These strategic needs include political voice, education, freedom from violence, access to personal income and a means of subsistence, and support for early childhood care. Women now confront several obstacles, such as pervasive poverty, a high rate of spousal abuse, and a lack of recognition for their contributions to the political, economic, and social spheres

3. Efforts made to support Women's Participation in STEM institutions in Southeast Asia

Table 3: Programmes to Improve the participation of Women in STEM in Southeast Asia

4. Intervention by the United Nations for the promotion of women in STEM fields

The United Nations is a central hub for dialogue and cooperation, facilitating collective solutions. Strong member state support provides a clear roadmap towards a shared goal addressing global challenges and caring for our typical global home.

On September 25, 2015, the United Nations General Assembly adopted the 2030 Agenda for Sustainable Development, succeeding the Millennium Development Goals (MDGs) as the global framework for international development. The Sustainable Development Goals to increase the participation of women in STEM are Goal 5, i.e., Achieve Gender equality and empower all women and girls, and Goal 10, i.e., Reduce inequality within and among countries. Goal 5 talks about the significant legal and legislative reforms essential to safeguard women's rights worldwide. Despite some progress, there remains a global gender wage gap, with women earning, on average, 24% less than men in the labor market. UN Women supports women in obtaining quality employment, acquiring assets, and exerting influence on institutions and public policies. It emphasizes the importance of acknowledging, reducing, and redistributing the burden of unpaid care. Goal 10 talks about how the wage gap varies between countries, with women earning 24% less than men. Additionally, they are also more prone to vulnerable employment, with up to 75% of women's jobs being informal or lacking protection in developing countries

Globally, 83% of domestic workers are women, and the majority lack legal entitlement to a minimum wage. Gender discrimination can intersect with other forms of discrimination, including age, disability, ethnicity, and economic status, compounding the burden of inequalities. UN Women actively addresses inequality within and among countries by advocating for decent work, social protection, and gender-sensitive economic policies globally. This includes eliminating discriminatory laws and practices and promoting appropriate legislation, policies, and actions. It advocates for employment policies that enhance labor market conditions and promote decent work for women, along with ensuring the safe migration and social protection of domestic workers

The aims and role of UNESCO for the promotion of women in STEM include enhancing the participation, achievement, and continuity of girls and women in STEM education and careers to narrow the gender gap in STEM professions, building the capacity of countries to provide gender-responsive STEM education, focusing on teacher training, educational content, and pedagogy, raising awareness about the significance of STEM education for girls and women. The UNESCO Engineering Initiative (UEI) focuses on increasing enrolment in engineering programs at the tertiary level to sustain and enhance socio-economic development, thereby encouraging and inspiring the youth to address contemporary challenges and ensure an adequate supply of engineers for future sustainable development. UEI provides a platform for youth to express their views on current engineering and development issues through events like the UNESCO Youth Forum and World Science Day 2013. Some more initiatives include the 8th UNESCO Youth Forum, World Science Day 2013, World Teachers' Day 2013, Engineering Outreach Event in Nigeria, and Airbus: Fly Your Ideas

STEM and Gender Advancement (SAGA) is a Global UNESCO project supported by Sida to reduce the gender gap in STEM across education and research levels. It identifies and addresses gaps in policy related to gender in science, technology, and innovation (STI) based on evidence, enhances capacity for data collection on gender in STEM fields, increases visibility, participation, and respect for women in STEM, improves tools for measuring the status of women and girls in science. Women in Science Report (UIS) data reveals that women comprise less than 30% of the world's researchers. The data also provides insights into the distribution of these women across public, private, and academic sectors, as well as their respective fields of research

Under L'Oréal-UNESCO For Women in Science, UNESCO and the L'Oréal Corporate Foundation aim to honor women researchers who have significantly contributed to addressing global challenges. By recognizing their work in areas such as new technologies, aging populations, and biodiversity threats, UNESCO and L'Oréal aim to acknowledge and support these women researchers, enabling them to continue their impactful contributions to science

The World Federation of Engineering Organizations (WFEO), established in 1968 under UNESCO, unites national engineering institutions from around 100 nations, representing over 30 million engineers. Its mission is to enhance engineering practice and advance the UN Sustainable Development Goals. The Paris Declaration (March 2018) reaffirms engineers' commitment to sustainable development, fostering socio-economic security, sustainable development, and global poverty alleviation through the proper application of technology. Top of Form. The WEF Future of Jobs Report aims to analyze and offer detailed insights into the extent of emerging trends by industry and geography. It focuses on the expected timeframe for these trends to impact job functions, employment levels, and required skills Top of Form

The Organization for Women in Science for the Developing World is an international organization (program unit of UNESCO) offering research training, career development, and networking opportunities for women scientists in the developing world at various career stages

5. Recommendations

Legislation plays a crucial role in establishing an equitable and equal workplace. Government policies effectively address the representation and rights of disadvantaged groups, fostering a fair and inclusive work environment. Schemes and measures should aim to mitigate disruptions and support the continuity of careers. Sensitization efforts should prioritize facilitating the return of women to the field of science after career breaks. There is a need to change the mindset of administrators and colleagues. Child-care gaps reflect a systemic bias against women's employment globally, and efforts need to be made to provide for child care at the workplace. This issue persists mainly in Western countries, hindering women's career paths as women in institutes face challenges balancing work and childcare due to the lack of support. Re-entering the profession is also hindered by age-based restrictions. The goal is to create workplaces that eliminate the need for career breaks. Institutions should actively recognize and reward excellence women demonstrate as more women will be encouraged to pursue STEM careers. Women may endure unacceptable behavior due to fear of potential repercussions. The most extreme form of social exclusion is sexual harassment. There is a need to underscore the importance of addressing such concerns promptly and thoughtfully. Sexual harassment encompasses instances where male scientists seek sexual favors from their juniors, along with various forms of verbal and other abusive behavior. Implementing effective measures for addressing and preventing harassment issues is the need of the hour. In this regard, gender sensitization programs for male and female scientists in major R&D institutes are facilitated by organizations such as UGC, CSIR, IAIR, ICMR, etc. Another concern is lack of proper transportation and safety, leading to lower stay-backs for work and informal interactions necessary for collaborations. Women are eight times more likely to reach leadership or supervisory positions if the organizational form of networking exists as opposed to hierarchy or bureaucracy in other disciplines. . Informal interactions between male and female researchers are crucial for dismantling organizational hierarchies and fostering a socio-cultural environment conducive to fruitful research. Top of Form Inspiration from seeing other women at the top in science fields inspires the ones in the lower positions too to achieve success. . Because of the trickle-down effect of few role models for women engineering students, there is a dearth of academic women engineers in visible leadership positions responsible for the under-representation of

women engineers in academia, industry and government. Encouraging women faculty to consider formal leadership positions and providing structured opportunities to explore those options will likely result in faculty members participating as academic leaders. . Family, gender ideology, and HR organizational practices determine the employment relationship and can help women in S&T; to grow in their careers. . Apart from the support from spouses and encouragement from peers, the workplace environment has to be more supportive of the needs of women, fewer differences remaining in interpersonal communication, and the adoption of specific strategies by women to work effectively in male-dominated organizations would be helpful . Family-friendly policies and organizational norms help women progress in their careers, whereas toxic work environments and societal issues inhibit the growth of women in the business sector. . Female scientist's productivity is not only positively influenced by intrinsic factors like self-motivation and confidence, but extrinsic institutional factors such as flexible working hours, woman-oriented management policies, fair appraisal, and mentorship also play an essential role (Fathima et al., 2020). Patriarchal norms restricting women's mobility hinder their effectiveness in career-related tasks and advocacy for career advancements. These barriers limit women's engagement in the public arena, impacting their roles as scientists. Top of Form To avoid discrimination in grant funding because of certain stereotypes of women being incapable of doing science, grants should be awarded based solely on the merit of proposals.

6. Conclusion

Limited research exists on barriers women face in STEM leadership roles, especially in Southeast Asian nations, despite extensive studies on women in leadership and STEM separately. The disparity in understanding the unique challenges women face in STEM leadership in these nations presents a significant barrier to advancing female participation and progression in STEM fields. It is imperative to explore these obstacles comprehensively, considering their potential impact on gender equity. Persistent barriers such as work-life balance struggles, gender bias, and microaggressions underscore the necessity for targeted research in this area. Support from individuals championing gender equality is crucial in enabling women to persist and advance in STEM leadership roles. Achieving gender equity requires a cultural transformation within STEM and leadership domains, fostering environments where women's achievements are valued and their authority is respected. There is a dire need to improve the chilly climate for women in STEM. This means revising institutional cultures to be more inclusive, creating educational environments that support and validate women's contributions, and implementing policies that promote equal treatment and support for women scientists, particularly from disadvantaged backgrounds. Addressing inequality in STEM participation necessitates educational and awareness programs, scholarship incentives, and flexible career policies to support women's advancement. Ensuring R&D; institutes provide essential facilities for women scientists, including appropriate toilets and transportation during odd hours, similar to provisions in call centers. They are encouraging professionals to realize their full potential, guiding them to available support, and promoting a professional approach to careers. Raising awareness about societal and institutional structures that facilitate successful navigation of family and career dynamics and promoting awareness and sensitivity across society and institutions regarding the challenges women face in successful participation in science.

References